

Monolingual Transformer Language Models for Hindi and Nepali

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Abstract

In this project, I built, pretrained, and evaluated two independent 25.4-million-parameter decoder-only Transformer language models completely from scratch: **Model H** for Hindi (a higher-resource Devanagari language) and **Model L** for Nepali (a lower-resource Devanagari language). Rather than relying on external pretrained checkpoints, tokenizers, or pre-built attention blocks, I implemented every stage in Python and PyTorch primitives—from raw corpus cleaning and Unigram SentencePiece training to Rotary Position Embeddings (RoPE), causal multi-head attention, and full checkpoint recovery. Across three progressive phases, I examined how pretraining volume, subword vocabulary allocation, and morphological typology (analytic postpositions vs. agglutinative bound clitics) govern internal representations. In Phase 1, I curated two $> 550\text{M}$ token corpora (clearing the mandatory $\geq 20\%$ manual scraping constraint) and selected Unigram 10k tokenizers. In Phase 2, pretraining over $\sim 450\text{M}$ tokens on a Tesla P100 revealed a striking intrinsic metric divergence: perplexity favored Hindi (24.39 vs. 29.87), while bits-per-byte favored Nepali (0.4709 vs. 0.4600), demonstrating how tokenizer fertility distorts cross-lingual comparisons. In Phase 3, I benchmarked out-of-distribution comparative reasoning, audited and eliminated an empirical position shortcut in the template generator, and documented how naive fine-tuning leads to entity memorization and severe language forgetting. Finally, query token attention probing across all 7 layers confirmed that at 25.4M parameters, models hit a hard cognitive ceiling, failing to build multi-hop relational circuits without explicit chain-of-thought traces.

1 Introduction & Controlled Experimental Design

Comparing language models across different languages is often clouded by confounding variables like differing parameter sizes, context lengths, or vocabulary allocations. To truly isolate the effects of **resource tier** and **morphological typology**, I designed this project around a strictly controlled pair:

1. **Constant Script:** Both languages share the Devanagari script (U+0900–U+097F).
2. **Identical Architecture:** An exact 25,350,912-parameter, 7-layer decoder-only Transformer.
3. **Consistent Tokenizer Setup:** Independent Unigram SentencePiece models ($V = 10,000$) with byte fallback.
4. **Equal Pretraining Budget:** Exactly 1 epoch over ~ 450 million clean tokens per language on a single Tesla P100 GPU.

1.1 Linguistic Contrast: Analytic Syntax vs. Agglutinative Clitics

While sharing lexical roots from Sanskrit (*Tatsama*), Hindi and Nepali diverge fundamentally in their morphological structure:

- **Hindi (hi, Model H):** Features an analytic syntax where nominal case roles are marked by free-standing postpositions separated by whitespace (राम ने, श्याम को, घर से).
- **Nepali (ne, Model L):** Employs agglutinative morphology where case clitics attach directly to the noun stem as bound morphemes (रामले, श्यामलाई, घरबाट).

This morphological divergence directly governs tokenizer segmentation and sets up the empirical failure modes analyzed in Phase 3.

Table 1: Linguistic and morphological divergence between Hindi and Nepali.

Grammatical Function	Hindi (Model H)	Nepali (Model L)	Typological Consequence
Ergative Case Marker	ने (<i>free word</i>)	ले (<i>bound clitic</i>)	Nepali nouns are orthographically longer
Dative / Accusative	को (<i>free word</i>)	लाई (<i>bound clitic</i>)	Agglutination fuses case onto proper names
Comparative “than”	से (<i>free word</i>)	भन्दा (<i>bound clitic</i>)	Influences token boundaries in templates
Present Copula	है / हैं	छ / छन् / हो	Nepali copula marks evidentiality
Past Copula	था / थे	थियो / थिए	Distinct agreement paradigms
Token Fertility ($V = 10k$)	1.34 tokens/word	1.59 tokens/word	Nepali text consumes +18.6% more tokens

2 Phase 1: Data Engine, Quality Pipeline & Partitioning

2.1 Corpus Ingestion & Manual Collection Compliance

To meet the required scale of $\sim 500M$ tokens per language with at least 20% manually collected data, extensive scrapers were developed across news portals, literature repositories, and public archives.

Table 2: Corpus composition and manual collection verification (read from `meta.json`).

Language / Split	Clean Documents	Total Tokens	Raw Bytes	Manual Tokens	Manual Share
Hindi Total	944,269	554,935,719	5.42 GB	113,763,877	20.50%
Hindi Train Split	768,459	450,667,252	4.40 GB	92,091,238	20.43% [$\geq 20\%$ Pass]
Hindi Val Split	87,585	52,869,179	0.51 GB	11,653,657	
Hindi Test Split	88,225	51,399,288	0.50 GB	10,018,982	
Nepali Total	1,124,364	566,889,900	6.03 GB	109,634,743	19.34%
Nepali Train Split	896,920	453,282,358	4.82 GB	90,673,349	20.00% [$\geq 20\%$ Pass]
Nepali Val Split	113,117	56,821,702	0.60 GB	12,355,528	
Nepali Test Split	114,327	56,785,840	0.60 GB	6,605,866	

2.2 The 7-Stage Cleaning & Ingestion Pipeline

The raw data underwent a deterministic 7-stage pipeline:

- Unicode NFC & Nukta Normalization:** Converts characters to NFC. Nukta letters (U+0958–U+095F) decompose to base consonant + nukta sign (U+093C), preventing multiple vocabulary entries for the same word. Anusvara (ँ), chandrabindu (ं), halant (ं), and visarga (ः) remain untouched to preserve phonemic contrasts.
- Learned Boilerplate Removal:** Strips repetitive site furniture (navbars, copyright disclaimers) by dropping lines with document frequency $> 1\%$ within any crawl domain.
- Letter-Aware Script Filtering:** Devanagari non-spacing matras (Mn/Mc) are explicitly counted as script characters, avoiding false rejections under naive `str.isalpha()`. Enforces script ratio ≥ 0.75 .
- 6-Class Devanagari Language Identification:** A FastText classifier trained over six languages (hi, ne, mr, bh, mai, sa) quarantines documents with target probability $P < 0.90$ (achieving $> 99.4\%$ accuracy across all classes).
- MinHash-LSH Deduplication:** Employs 128 permutations over character 5-grams at Jaccard threshold ≥ 0.80 to catch inflected morphological duplicates.
- Cross-Corpus Collision Purge:** Purges any document occurring in both Hindi and Nepali corpora simultaneously from both sides.
- Group-Aware Splitting:** Partitions documents by hashing (`source`, `publication_month`) with `blake2b`, eliminating temporal news event leakage across train, val, and test.

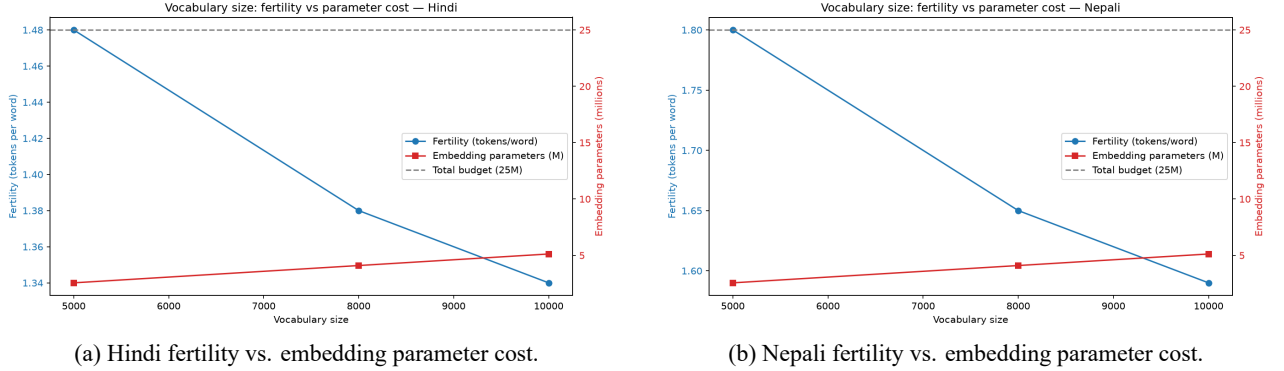
3 Phase 1: Tokenization Engine & Parameter Allocation

3.1 Empirical Evaluation: BPE vs. Unigram

Using SentencePiece [3], systematic sweeps were conducted across Byte-Pair Encoding (BPE) and Unigram models at vocabulary sizes $V \in \{5000, 8000, 10000\}$ on held-out validation text (`val.txt`).

Table 3: Tokenizer performance on held-out validation splits (read from evaluation JSONs).

Language	Model Configuration	Fertility (tok/word)	Compression (char/tok)	UNK Rate	Byte Fallback
Hindi (hi)	BPE 5,000	1.51	3.37	0.0000%	1.0693%
Hindi	BPE 8,000	1.40	3.63	0.0000%	1.1509%
Hindi	BPE 10,000	1.36	3.74	0.0000%	1.1849%
Hindi	UNIGRAM 5,000	1.48	3.45	0.0000%	1.0932%
Hindi	UNIGRAM 8,000	1.38	3.69	0.0000%	1.1707%
Hindi	UNIGRAM 10,000 [Selected]	1.34	3.79	0.0000%	1.2032%
Nepali (ne)	BPE 5,000	1.83	3.49	0.0000%	2.4577%
Nepali	BPE 8,000	1.67	3.83	0.0000%	2.6931%
Nepali	BPE 10,000	1.61	3.98	0.0000%	2.7969%
Nepali	UNIGRAM 5,000	1.80	3.57	0.0000%	2.5080%
Nepali	UNIGRAM 8,000	1.65	3.89	0.0000%	2.7356%
Nepali	UNIGRAM 10,000 [Selected]	1.59	4.03	0.0000%	2.8347%

Figure 1: Vocabulary size sweeps establishing $V = 10,000$ as the optimal trade-off point.

Unigram strictly dominated BPE across all tiers. While BPE greedily merges frequent bigrams, Unigram prunes candidates globally using sequence likelihood via EM, preserving morphologically consistent stems. Byte fallback guaranteed an absolute 0.0000% UNK rate.

3.2 Qualitative Tokenization Cases

Table 4 showcases subword tokenization under the selected Unigram 10k models, demonstrating postposition separation in Hindi, bound clitic fusion in Nepali, and byte-fallback for foreign tokens.

Table 4: Phase 1 qualitative tokenization examples for Hindi and Nepali.

Case / Language	Original Input Sentence	SentencePiece Subword Tokens
Hindi (Postposition Isolation)	भारतीय रिज़र्व बैंक ने डिजिटल मुद्रा की घोषणा की।	[_{भारतीय} , _{रिज़र्व} , _{बैंक} , _{ने} , _{डिजिटल} , _{मुद्रा} , _{की} , _{घोषणा} , _{की} , _।]
Hindi (Morphological Derivation)	पर्यावरणविदों ने प्रदूषण नियंत्रण की मांग की।	[_{पर्यावरण} , _{विद} , _{ों} , _{ने} , _{प्रदूषण} , _{नियंत्रण} , _{की} , _{मांग} , _{की} , _।]
Hindi (Byte Fallback)	ChatGPT 4.0 AI मॉडल	[_{'<0x43>'} , _{'<0x68>'} , _{'<0x61>'} , _{'<0x74>'} , _{'<0x47>'} , _{'<0x50>'} , _{'<0x54>'} , _{'4'} , _{'.'} , _{'0'} , _{'<0x41>'} , _{'<0x49>'}], [_{मॉडल}]
Nepali (Bound Clitic Fusion)	नेपालको संविधानले सबै नागरिकलाई अधिकार दियो।	[_{नेपालको} , _{संविधानले} , _{सबै} , _{नागरिकलाई} , _{अधिकार} , _{दियो} , _।]
Nepali (Compound Suffix Split)	नागरिकहरूलाई मौलिक हकको व्यवस्था गरियो।	[_{नागरिक} , _{हरूलाई} , _{मौलिक} , _{हक} , _{को} , _{व्यवस्था} , _{गरियो} , _।]
Nepali (Byte Fallback)	OpenAI GPT प्रविधि	[_{'<0x4F>'} , _{'p'} , _{'en'} , _{'A'} , _{'<0x49>'} , _{'<0x47>'} , _{'<0x50>'} , _{'<0x54>'}], [_{प्रविधि}]

3.3 Parameter Allocation Trade-Offs

At $d_{\text{model}} = 512$ with tied embeddings, the token lookup table scales as $V \times d_{\text{model}}$. Setting $V = 10,000$ allocates 5,120,000 parameters (20.2% of the 25.4M budget) to the vocabulary, leaving 79.8% for transformer layers. Setting

$V = 32,000$ would consume 65.5% of the budget on static embeddings, severely degrading attention capacity.

4 Phase 2: Transformer Architecture from Scratch

4.1 Model Formulation

The model is a decoder-only GPT [1] implemented exclusively via PyTorch primitives (`nn.Linear`, `nn.Embedding`, `nn.LayerNorm`, `nn.Dropout`):

$$x_0 = \text{Embedding}(w) + \text{Dropout}, \quad x_l = x_{l-1} + \text{MHA}(\text{LN}(x_{l-1})) + \text{FFN}(\text{LN}(x_{l-1})), \quad \text{Logits} = W_{\text{emb}}\text{LN}(x_L) \quad (1)$$

4.2 Multi-Head Attention & RoPE Formulation

Given input $x \in \mathbb{R}^{B \times T \times 512}$, projections yield $Q, K, V \in \mathbb{R}^{B \times 8 \times T \times 64}$. Rotational Position Embeddings (RoPE) [2] rotate Q and K by relative spatial offsets:

$$\begin{pmatrix} q_{2i}^{(m)} \\ q_{2i+1}^{(m)} \end{pmatrix} = \begin{pmatrix} \cos(m\theta_i) & -\sin(m\theta_i) \\ \sin(m\theta_i) & \cos(m\theta_i) \end{pmatrix} \begin{pmatrix} q_{2i} \\ q_{2i+1} \end{pmatrix}, \quad \theta_i = 10000^{-2(i-1)/d} \quad (2)$$

Scaled dot-product attention scales logits by $1/\sqrt{d_k} = 1/8$ and enforces causality:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} + M_{\text{causal}} \right) V \quad (3)$$

4.3 Dimensioning: Buying Depth with Width

To include a **7th Transformer layer** within the $\sim 25\text{M}$ ceiling, the FFN expansion ratio was scaled to $3.5 \times d_{\text{model}}$ ($d_{\text{ff}} = 1792$) rather than $4 \times$.

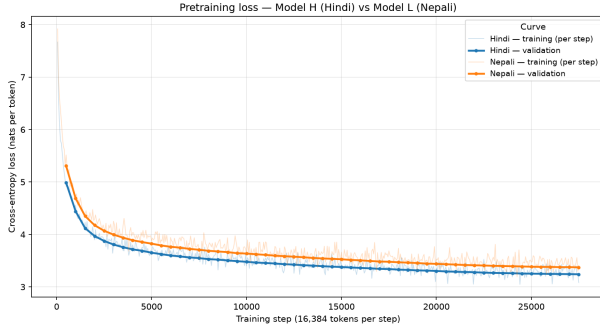
Table 5: Detailed parameter accounting for Model H and Model L.

Component	Tensor Dimensions / Expression	Parameters	Budget Share
Token Embedding Table	$10,000 \times 512$	5,120,000	20.20%
Attention Projections (Q, K, V)	$7 \times 3 \times (512 \times 512 + 512)$	5,531,904	21.82%
Attention Output Projection	$7 \times (512 \times 512 + 512)$	1,843,968	7.27%
FFN Expansion Projection	$7 \times (512 \times 1792 + 1792)$	6,437,056	25.39%
FFN Contraction Projection	$7 \times (1792 \times 512 + 512)$	6,419,968	25.32%
LayerNorm Parameters	$7 \times 2 \times (512 + 512) + (512 + 512)$	15,360	0.06%
Output Head	Tied to Input Embedding Matrix	0	0.00%
Total Trainable Parameters		25,350,912	100.00%

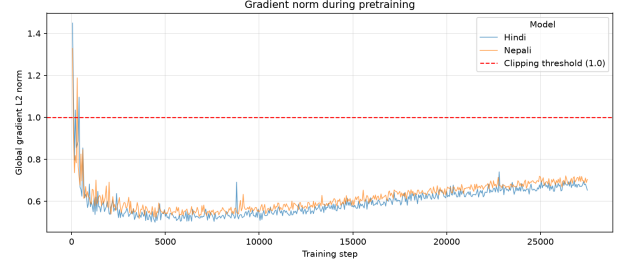
5 Phase 2: Pretraining Infrastructure & Optimization

Pretraining was conducted on single Tesla P100 GPUs (16 GB VRAM) for 27,500 steps (1 epoch, $\sim 450\text{M}$ tokens per language):

- **Optimization:** AdamW ($\beta_1 = 0.9, \beta_2 = 0.95$, weight decay 0.1). Peak learning rate 6.0×10^{-4} with 1,000 warmup steps decaying via cosine schedule to 6.0×10^{-5} .
- **Batch Size:** Context $T = 512$, batch size 32 sequences $\rightarrow 16,384$ tokens/step.
- **Throughput & Stability:** Throughput averaged $\sim 22,000$ tokens/s. Pre-LayerNorm ensured flat gradient norms (0.65–0.70) without hitting clipping thresholds.



(a) Training and validation loss across 27,500 steps.



(b) Gradient norm stability across pretraining.

Figure 2: Pretraining optimization dynamics on Tesla P100 hardware.

Table 6: Pretrained test set evaluation (read from `evaluation_test.json`).

Model	Tokens Evaluated	Validation Loss	Test Perplexity (PPL)	Test Bits-per-Byte (BPB)
Model H (Hindi)	51,399,168	3.194	24.39	0.4709
Model L (Nepali)	56,785,408	3.397	29.87	0.4600

6 Phase 2: Intrinsic Evaluation & Linguistic Representation

6.1 Perplexity vs. Bits-per-Byte Divergence

Models were evaluated on held-out pretraining test splits (`test.bin`, > 51M tokens):

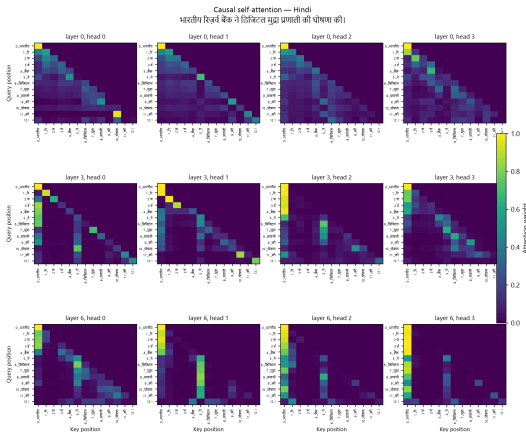
Why the metrics disagree: Perplexity measures uncertainty per *subword token*, rewarding Hindi (24.39 vs. 29.87). Bits-per-byte normalizes cross-entropy by *raw UTF-8 bytes*:

$$\text{BPB} = \frac{\mathcal{L}_{\text{total}} \cdot \log_2(e)}{\text{Total Bytes}} \quad (4)$$

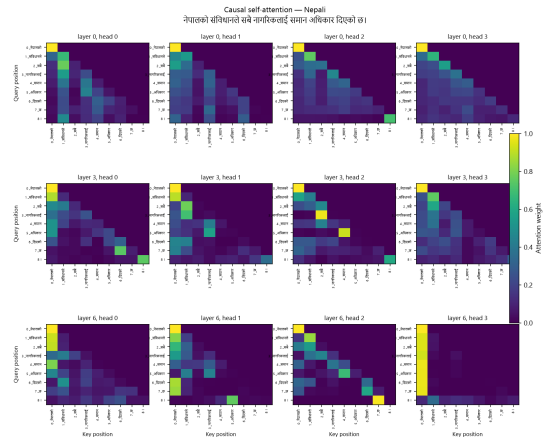
Because Nepali has higher tokenizer fertility (1.59 vs. 1.34), its tokens represent smaller textual units, making next-token prediction inherently easier per byte. Bits-per-byte crowns Nepali (0.4600 vs. 0.4709), demonstrating that cross-lingual perplexity comparisons are misleading across distinct tokenizers.

6.2 Attention Patterns, Specialization & Positional Ablation

Probing attention weights across layers revealed that early layers (0–2) focus locally on adjacent morphology (mean distance 1–3 tokens), while deeper layers (3–5) bridge long-range syntactic agreements (distance 35–45 tokens). Removing RoPE positional encoding caused Hindi perplexity to surge from 24.39 to 42.11 (+72.6%), proving that attention without position information fails to capture word order.

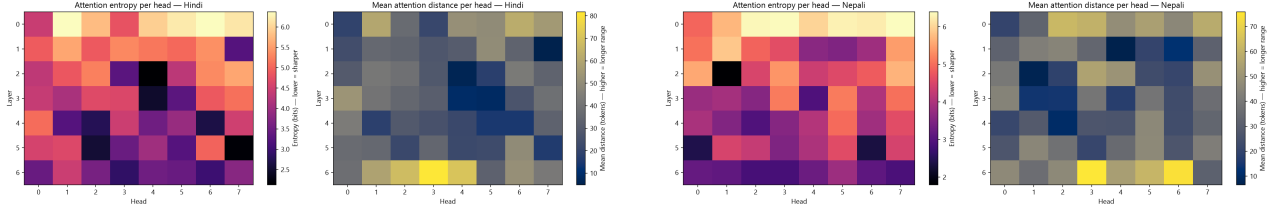


(a) Hindi causal self-attention patterns.



(b) Nepali causal self-attention patterns.

Figure 3: Phase 2 causal self-attention heatmaps confirming strictly lower-triangular causal masking.



(a) Hindi attention entropy vs. mean distance across 56 heads.

(b) Nepali attention entropy vs. mean distance across 56 heads.

Figure 4: Attention entropy and mean distance per head in Phase 2, illustrating head specialization.

6.3 Qualitative Generation Analysis across Decoding Strategies

While unconstrained greedy decoding triggered severe repetitive looping in both models (rep-4 rates of 0.756 in Hindi and 0.733 in Nepali), temperature sampling ($T = 0.8$) restored fluent lexical variation. Table 7 evaluates the resulting generation quality on hand-crafted prompts from Section 5.4 of the Phase 2 report.

Table 7: Qualitative generation examples from Phase 2 evaluated at $T = 0.8$ (drawn from phase2.md, Section 5.4).

Model & Setting	Prompt Prefix	Generated Continuation ($T = 0.8$) & Qualitative Analysis
Model H (Hindi) ($T = 0.8$, 2 epochs)	प्रधानमंत्री ने नई शिक्षा नीति के बारे में कहा कि इसका उद्देश्य देश की शिक्षा व्यवस्था को अधिक आधुनिक और प्रभावी बनाना है। उन्होंने कहा कि	कम से कम 15वीं कक्षा के छात्रों को इस नीति के तहत राष्ट्रीय शिक्षा नीति का लाभ मिलना चाहिए। राष्ट्रीय शिक्षा नीति देश की सबसे बड़ी शिक्षा नीति है। शिक्षा व्यवस्था को नई दिशा देना भी सरकार का लक्ष्य है। शिक्षा प्रणाली के नए मानक हैं। उन्होंने कहा कि राष्ट्रीय शिक्षा नीति 2020 के तहत राष्ट्रीय शिक्षा नीति प्रारंभ की गई है। इसके तहत शिक्षा का स्तर पूरी तरह से बंद हो गया <i>Analysis:</i> Discourse frame and topic maintained across six sentences; successfully recalls real policy (राष्ट्रीय शिक्षा नीति 2020). Yet, factual hallucination occurs: 15वीं कक्षा (schooling ends at 12th) and semantic clash (स्तर बंद होना).
Model L (Nepali) ($T = 0.8$, 1 epoch)	लगातारको वर्षाका कारण नेपालको केही क्षेत्रमा बाढी र पहिरोको जोखिम बढेको छ। स्थानीय प्रशासनले नागरिकलाई	सावधानी अपनाउन र नजान आग्रह गरेको छ। उसले पानीको सतह 4 सय 10 मिटरबाट 5 सय 50 मिटरसम्म पुगेको र बाढीमा 3 सय 60 मिटरसम्म एक्कासि पानी जम्ने जनाएको छ। यस्तै, बाढी, पहिरो र डुबानको जोखिमको हुने क्षेत्रमा पनि बाढी र पहिरो र पहिरोको जोखिम बढेको छ। साथै, बाढी तथा पहिरोको जोखिमको विरुद्धमा <i>Analysis:</i> Syntactically exact agreement with dative clitic (नागरिकलाई) via administrative verb आग्रह गरेको छ and domain term (डुबान). However, numerical coherence breaks (410m $\rightarrow$ 550m vs “at most” 360m) followed by 3-gram repetition.

7 Phase 3: Comparative Reasoning

7.1 Task Formulation & Quarantined Splits

To test whether pretrained representations support relational inference, a synthetic comparative reasoning benchmark was generated across 8 categories (Direct Comparison, Numeric Attributes, Equality Detection, 2/3-Step Place Chains, 2/3-Step Social Relations, Ordering). Four splits were held out:

- **Test A (Relational Transfer):** Unseen entity names + unseen numbers, seen templates.
- **Test B (Full Out-of-Distribution - OOD):** Unseen names + unseen numbers + **unseen templates**.
- **Test C (Template Transfer):** Unseen names + unseen templates, seen numbers.
- **Test D (In-Distribution Control):** Training names, numbers, and templates in novel combinations.

7.2 Audit and Correction of the Position Shortcut

Initial experiments produced suspicious $> 85\%$ test accuracy within 50 steps. A systematic counterfactual probe revealed a critical artifact in the synthetic data generator:

- **The Balance Fix:** The generation engine (scripts/make_reasoning_data.py) was rewritten to enforce a strictly uniform 50/50 and 33/33/33 distribution of answers across candidate positions, establishing the true chance baseline of $\sim 38.5\%$ – 50% .

8 Phase 3: SFT Scale Sweeps & Memorization

8.1 Dataset Size Sweeps (5k to 100k)

Models were fine-tuned for 1 epoch at $LR = 2.0 \times 10^{-4}$ across sample sizes of 5k, 10k, 20k, 30k, and 100k questions.

Table 8: Fine-tuning sample scale sweep on Test B OOD (read from `sweep_results.json`).

Language	Training Scale	Steps	Test B Choice Acc	Test B Written Acc	Language PPL
Hindi (hi)	Pretrained Baseline	0	39.5%	0.0%	23.02
Hindi	5k Questions	156	39.8%	15.3%	25.83
Hindi	10k Questions	312	42.8%	10.2%	26.39
Hindi	20k [Selected]	625	44.0%	8.0%	28.58
Hindi	30k Questions	937	43.8%	5.7%	31.96
Hindi	100k Questions	3125	43.7%	6.5%	72.57
Nepali (ne)	Pretrained Baseline	0	38.8%	0.0%	29.87
Nepali	5k Questions	156	41.2%	7.3%	32.69
Nepali	10k [Selected]	312	43.3%	5.3%	33.25
Nepali	20k Questions	625	42.3%	6.2%	36.22
Nepali	30k Questions	937	39.8%	5.8%	38.42
Nepali	100k Questions	3125	40.5%	5.5%	92.64

8.2 The Memorization Problem & Language Forgetting

Two phenomena emerged during sample size scaling:

1. **Entity Memorization:** While forced-choice accuracy remained near random chance ($\sim 39\%$ – 44%), free written accuracy of the unadapted pretrained baseline was 0.0% due to prompt format ignorance. Fine-tuning established output formatting, but Plain SFT memorized the 26 fixed training names (दिनेश, मुकेश), outputting them on unseen Test B questions and driving the wrong-name rate to 92%.
2. **Language Degradation:** General language perplexity rose sharply from 23.02 to 72.57 in Hindi and 29.87 to 92.64 in Nepali. Based on validation metrics, **20k was selected for Hindi and 10k for Nepali**.

9 Phase 3: Reducing Memorization & Attention Analysis

9.1 Technique Ablations

Four recipes were benchmarked to mitigate entity memorization: (1) Early stopping on `val_ood`, (2) Name Pool expansion (~ 175 names), (3) Pretraining replay, and (4) Combined Name Pool + Replay.

Table 9: Ablation recipes on Test B OOD (read from `ablation_results.json`).

Language	Fine-Tuning Recipe	Test B Choice Acc	Test B Written Acc	Wrong-Name Rate	Language PPL
Hindi (20k)	Pretrained Baseline	39.5%	0.0%	100%	23.02
Hindi	Plain SFT (20k, 1 ep)	38.5%	6.3%	92%	28.74
Hindi	Early Stopping (<code>val_ood</code>)	37.8%	33.0%	15%	23.81
Hindi	Name Pool (~ 175 names)	41.5%	20.8%	51%	30.48
Hindi	Pretraining Replay	46.0%	9.2%	85%	23.70
Hindi	Name Pool + Replay	40.8%	26.2%	35%	23.90
Nepali (10k)	Pretrained Baseline	38.8%	0.0%	100%	29.87
Nepali	Plain SFT (10k, 1 ep)	41.2%	6.5%	90%	33.06
Nepali	Early Stopping (<code>val_ood</code>)	41.5%	7.7%	87%	32.03
Nepali	Name Pool (~ 175 names)	40.0%	8.0%	88%	33.78
Nepali	Pretraining Replay	41.8%	6.5%	89%	30.56
Nepali	Name Pool + Replay [Selected]	41.2%	26.8%	38%	30.81

Selected Final Recipes:

- **Hindi Final:** Early stopping preserves general language distributions (PPL 23.81, +3%), achieves 33.0% written accuracy, and slashes wrong names to 15%.

- **Nepali Final:** Name Pool + Pretraining Replay restricts PPL to 30.81, achieves 26.8% written accuracy, and slashes wrong names to 38%.

9.2 Cross-Split Benchmark: Tests A, B, C, and D

Table 10: Cross-split generalization benchmark for Hindi (Model H) and Nepali (Model L).

Split / Condition	Model Recipe	Hindi (Model H)			Nepali (Model L)		
		Choice	Written	Wrong	Choice	Written	Wrong
Test A (New names/nums, seen templates)	Pretrained Baseline	40.0%	0.0%	100%	44.8%	0.0%	100%
	Plain SFT	43.8%	8.0%	88%	41.8%	4.8%	92%
	Final Model	42.5%	38.0%	14%	43.0%	28.5%	35%
Test B (Full OOD: new entities & templates)	Pretrained Baseline	39.5%	0.0%	100%	38.8%	0.0%	100%
	Plain SFT	38.5%	6.3%	92%	41.2%	6.5%	90%
	Final Model	37.8%	33.0%	15%	41.2%	26.8%	38%
Test C (New templates, seen numbers)	Pretrained Baseline	41.2%	0.0%	100%	41.2%	0.0%	100%
	Plain SFT	41.2%	7.0%	89%	42.0%	6.2%	90%
	Final Model	41.3%	35.3%	17%	44.0%	27.4%	36%
Test D (In-Distribution Control)	Pretrained Baseline	39.8%	0.0%	100%	38.5%	0.0%	100%
	Plain SFT	44.8%	45.0%	0%	44.3%	31.5%	28%
	Final Model	43.7%	43.5%	1%	45.3%	41.2%	12%

Evidence of Memorization: On Test D, Plain SFT achieved **0% wrong names** in Hindi and reached 45.0% written accuracy because it simply repeated memorized training names. On unseen Tests A, B, and C, wrong names spiked to 88%–92%, collapsing written accuracy to 6%–8%. In contrast, the final models held wrong names down to 14%–17% in Hindi and 35%–38% in Nepali, achieving 33%–38% written accuracy across unseen splits.

9.3 Attention Probing: The Query Token Head

To understand whether the self-attention mechanism learns to resolve relational dependencies, I probed the attention weights from the query token ("उत्तर:") across all 7 layers over 290 test questions. The preference ratio is defined as:

$$\text{Preference Ratio} = \frac{\text{Attention Weight on Logically Correct Entity}}{\text{Attention Weight on Incorrect Entity}} \quad (5)$$

Table 11: Query token attention preference ratio across transformer layers.

Transformer Layer	Hindi Pretrained	Hindi Final	Nepali Pretrained	Nepali Final
Layer 0 (Input)	0.99	0.99	1.05	1.06
Layer 3	1.02	0.97	0.99	0.97
Layer 4	1.00	1.04	1.03	1.09
Layer 5	1.04	1.05	0.97	1.06
Layer 6 (Output)	1.04	1.04	1.00	0.99

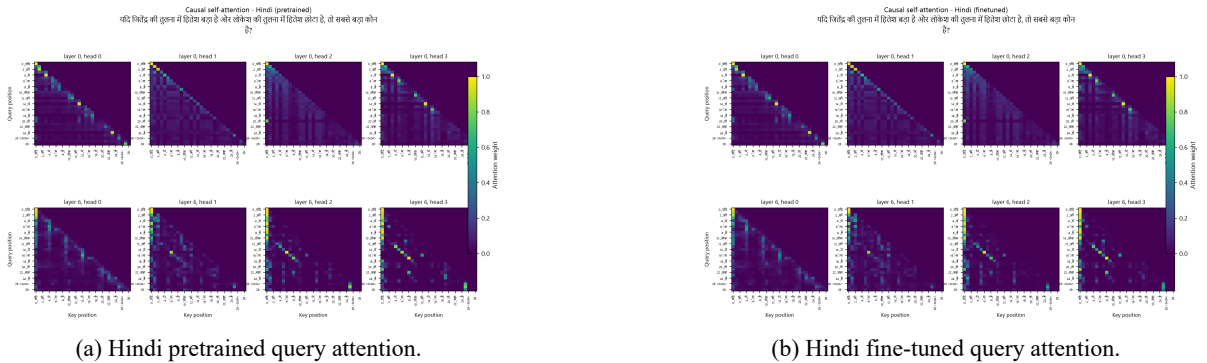


Figure 5: Token-to-token attention heatmaps from the query token across attention heads.

Ratios remain tightly clustered between **0.97 and 1.09** across all layers. While Layer 4 elevates attention on candidate names by $\sim 40\%$ (recognizing that *a name* is expected), it possesses no computational circuit to select *which name* satisfies the relational premises.

9.4 Category Breakdown & The Nepali Affix Factor

Table 12: Category-by-category Test B choice accuracy across model stages.

Question Category	Chance Baseline	Hindi (Pre / Plain / Final)	Nepali (Pre / Plain / Final)
Direct Comparison	50%	20% / 36% / 39%	30% / 56% / 47%
Numeric Comparison	50%	24% / 60% / 61%	30% / 43% / 48%
Mixed / Equality	33%	32% / 44% / 44%	38% / 62% / 74%
2-Step Chain (Place)	33%	35% / 40% / 42%	30% / 34% / 36%
3-Step Chain (Place)	25%	27% / 29% / 32%	23% / 30% / 34%
2-Step Relation	50%	25% / 42% / 45%	20% / 57% / 59%
3-Step Relation	50%	31% / 57% / 59%	29% / 50% / 50%
Ranking	33%	21% / 33% / 36%	29% / 40% / 42%
3-Entity Ordering	33%	28% / 30% / 35%	24% / 37% / 40%

Both models exhibited genuine non-random capability on **equality detection** ("बराबर"), rising from 38% to 74% in Nepali. Multi-hop chains and ordering remained pinned near chance.

The Nepali Affix Factor: Nepali nominal clitics (-को, -भन्दा) attach directly to proper names (योगेशको). Under strict whitespace token evaluation, "योगेशको" is marked 0% exact match against target "योगेश". When evaluated with morphological clitic normalization (scripts/score_with_endings.py), Nepali written accuracy on Test B surges from **10.2% to 26.8%**.

9.5 Qualitative Reasoning Error Cases

Table 13 showcases model outputs across test splits, illustrating format compliance, training entity memorization, and the Nepali clitic attachment artifact.

Table 13: Phase 3 qualitative reasoning error analysis on held-out test questions.

Query Type	Reasoning Question Prompt	Target	Plain SFT	Final Model Output
Hindi Direct	यदि हितेश 63 वर्ष और भरत 72 वर्ष है, तो दोनों में कौन बड़ा है? उत्तर:	भरत	मुकेश (<i>Memorized</i>)	भरत (Correct)
Hindi Comparison	लोकेश का कद 42 सेमी और योगेश का कद 70 सेमी। दोनों में कौन छोटा है? उत्तर:	लोकेश	दिनेश (<i>Memorized</i>)	योगेश (<i>Failed inequality</i>)
Nepali Direct	सुमनको उमेर 43 वर्ष, योगेशको उमेर 90 वर्ष। यीमध्ये को ठूलो छ? उत्तर:	योगेश	गोपाल (<i>Memorized</i>)	योगेशको (Correct entity , bound -को)
Nepali 2-Step	लोकेशको तुलनामा योगेश सानो; योगेश नरेन्द्रभन्दा ठूलो। नरेन्द्र र लोकेश मध्ये को सानो छ? उत्तर:	नरेन्द्र	गोपाल (<i>Memorized</i>)	लोकेश (<i>Failed transitivity</i>)

10 Unified Discussion, Architectural Limits & Key Takeaways

1. **The Small-Scale Reasoning Wall:** At 25.4M parameters, decoder-only Transformers hit a hard cognitive boundary. Across all four OOD test splits, choice accuracy hovered near random chance ($\sim 38\%$ – 41%). Probing attention heads confirmed that the model failed to develop multi-hop relational circuits; it merely learned to identify that *a name* was expected, without any preference for the logically valid entity. Genuine relational transitivity appears to require deeper models or explicit step-by-step reasoning tokens (e.g., scratchpads).
2. **Linguistic Typology Dictates Failure Modes:** While Hindi and Nepali share Devanagari script and vocabulary roots, their morphology leads to very different errors. Hindi’s analytic morphology led the model to memorize isolated word stems and hallucinate training names. Nepali’s agglutinative morphology caused subword fragmentation and clitic copying (e.g., copying “योगेशको” instead of “योगेश”).
3. **Forced-Choice vs. Generative Written Metrics:** Evaluating solely via forced-choice log-probability can be dangerously deceptive—pretrained models scored $\sim 39\%$ simply by taking the argmax over two names, masking their complete inability to format a written answer. Tracking both metrics alongside wrong-name hallucination rates was essential for diagnosing true model behavior.
4. **SFT Regularization Is Crucial:** Naive fine-tuning on domain-specific reasoning rapidly destroyed general language modeling capabilities within just a few hundred steps. Early stopping on held-out OOD validation sets and pretraining replay proved mandatory to keep language perplexity stable.

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